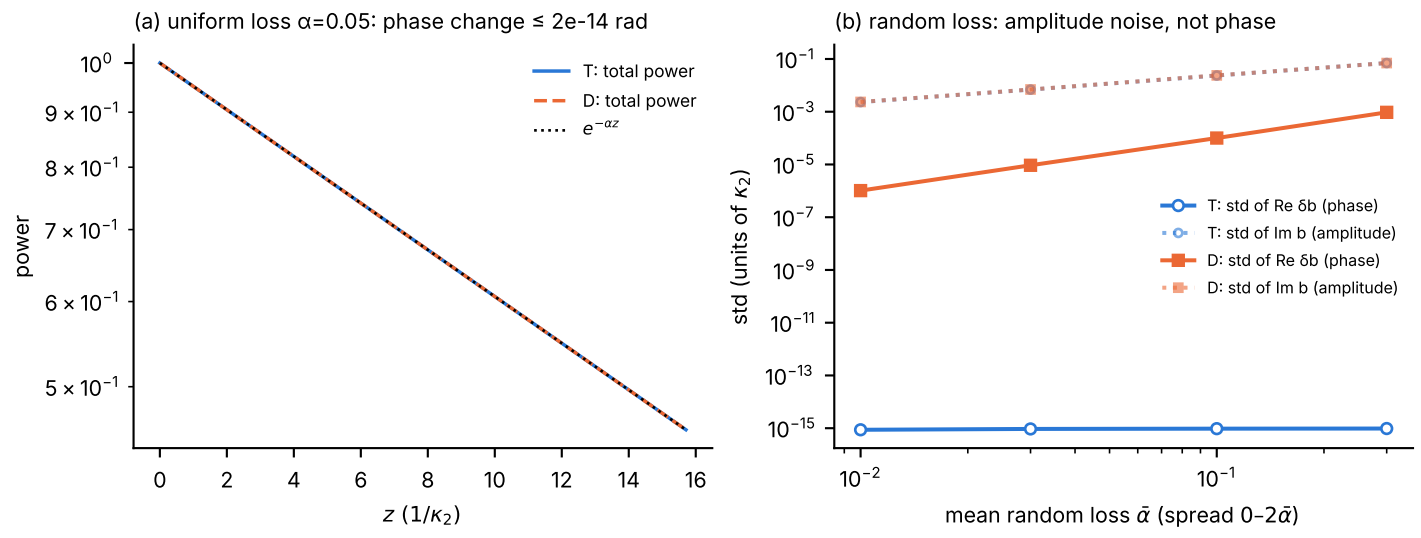
(a) uniform loss α=0.05: phase change ≤ 2e-14 rad
T: total power
D: total power
$e^{-\alpha z}$
power
10^{0}
9×10^{-1}
8×10^{-1}
7×10^{-1}
6×10^{-1}
5×10^{-1}
0
2
4
6
8
10
12
14
16
z ($1/\kappa_2$)
(b) random loss: amplitude noise, not phase
std (units of κ_2)
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-15}
T: std of Re δb (phase)
T: std of Im b (amplitude)
D: std of Re δb (phase)
D: std of Im b (amplitude)
10^{-2}
10^{-1}
mean random loss $\bar{\alpha}$ (spread 0–2$\bar{\alpha}$)